SOUMYADIP SEN

Full Stack Developer

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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on freelance experience building responsive, data-driven web applications using React.js, Node.js, Express.js, and SQL/NoSQL databases. Skilled at end-to-end project delivery — from client requirements gathering to deployment and maintenance — combined with a strong foundation in AI/ML, IoT integration, and cloud-connected systems. Proven track record of independently shipping multiple full-stack and IoT-based projects.

EXPERIENCE

Freelance Full Stack Developer — Self-employed *Feb 2026 – Present*

- Designed and developed responsive, client-facing web applications using HTML, CSS, JavaScript, and React.js.
- Built and integrated back-end functionality with Node.js, Express.js, and MySQL/MongoDB to support dynamic, data-driven websites.
- Collaborated directly with clients to gather requirements, scope projects, and deliver features on schedule.
- Deployed and maintained production websites, resolving bugs and implementing performance improvements.
- Managed end-to-end project delivery independently, using Git and GitHub for version control.

Freelance Tutor — Self-employed *2022 – Present*

- Developed structured lesson plans and assessments in PowerPoint and Word to support student learning outcomes.
- Designed and delivered presentations and reports for training and marketing purposes.
- Maintained academic records and parent communication through organized Excel tracking sheets.
- Created and proofread educational content on technology and career guidance topics.
- Taught 20+ students Mathematics, Science, and other subjects up to Class 10, improving academic performance.

EDUCATION

Adamas University *Sep 2024 – Jun 2028*

Bachelor of Technology (B.Tech) – Computer Science & Engineering, CGPA: 8.23 / 10

PROJECTS

Heartery – IoT-Based Smart Heart Health Monitoring System — *ESP32 • AD8232 ECG • MAX30102 • ThingSpeak/Firebase*

- Developed a low-cost, portable IoT-based cardiac monitoring system using ESP32, AD8232 ECG, and MAX30102 pulse-oximetry sensors to monitor ECG, heart rate, and SpO_2 in real time.
- Implemented signal acquisition, digital filtering, PQRST waveform visualization, and cloud connectivity using ThingSpeak/Firebase, with OLED-based real-time health-data display.
- Built and calibrated a functional prototype achieving 97.5% heart-rate and 98.9% SpO_2 accuracy against reference devices, with scope for remote monitoring and AI-based cardiac anomaly detection.

Fall Detection System for Elderly with Online Dashboard

- Built a real-time fall detection system to enhance safety monitoring for elderly individuals living independently.
- Designed sensor-based detection logic to accurately identify falls and trigger instant alerts to caregivers.
- Developed a web-based online dashboard for caregivers to remotely monitor activity status and view alert history.
- Integrated backend data pipelines to log and visualize health and movement patterns for informed care decisions.

DHRUVA – Polar Science Knowledge & Outreach Portal — *Integrated Knowledge Repository & Media Dissemination Platform*

- Developed a centralized digital platform for organizing and disseminating polar research, including expedition reports, scientific datasets, publications, photographs, videos, and institutional activities.
- Implemented AI-assisted content generation to transform scientific information into accessible website and social-media content, improving research outreach and knowledge dissemination.
- Designed a structured repository with secure access, metadata-based organization, search/discovery, and scalable architecture to support scientists, researchers, students, and the public.

PathVistā – AI-Powered Career & Education Navigator — *React.js • Node.js/Express • PostgreSQL • AI/ML • Maps API • Firebase*

- Built a personalized career guidance platform that maps student aptitude and interests to suitable career paths, courses, and government colleges.
- Implemented location-based college discovery, scholarship matching, eligibility/cut-off information, career roadmaps, multilingual support, and offline-first access.
- Designed the system to improve career clarity and educational accessibility for students, particularly those from rural and low-connectivity regions.

GyanBhandar – Digital eBook Management System

- Developed a digital eBook management platform for efficiently organizing, searching, categorizing, and accessing a centralized collection of electronic books.
- Implemented features for eBook uploading, metadata management, category-based browsing, search, user access, and digital reading.
- Designed a user-friendly system that simplifies digital library management and provides structured access to educational and reading resources.

WhereIs – Digital Bus Tracking & Navigation System

- Developed a digital bus tracking platform that enables users to locate and track buses in real time based on bus type, route, and availability.
- Implemented route-based bus discovery, live location tracking, estimated arrival information, and an intuitive interface to simplify daily public transportation planning.
- Designed the system to improve commuter convenience, route visibility, and transportation efficiency through centralized digital tracking.

AI Logic Theorem Solver

- Developed an AI-powered system to automatically parse, evaluate, and solve formal logic theorems and propositional expressions.
- Implemented rule-based and AI-assisted inference techniques to validate logical statements and generate step-by-step proofs.
- Designed an intuitive interface for users to input logical expressions and view solution steps in real time.
- Improved computational efficiency of the solving engine through optimized algorithm design in Python.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL

Full Stack Development: React.js, Node.js, Express.js, HTML, CSS, JavaScript, MySQL, MongoDB, PostgreSQL, REST APIs, Firebase

Data Science & AI: Machine Learning, Data Analysis, Predictive Analytics, Generative AI, Large Language Models (LLMs), Synthetic Data Generation, Prompt Engineering

Tools & Technologies: Pandas, NumPy, DuckDB, ChromaDB, FastAPI, Git, GitHub

Core Competencies: Statistics, Data Preprocessing, Data Visualization, Model Evaluation, Differential Privacy, Responsible AI, Compliance-Aware AI Systems

CERTIFICATIONS

- Data Science and Analytics – HP LIFE
- Digital Business Skills – HP LIFE
- Career Essentials in Generative AI – Microsoft & LinkedIn
- Build Your Generative AI Productivity Skills with Microsoft & LinkedIn
- Generative AI Tools – NASSCOM

ACHIEVEMENTS

- Independently delivered multiple freelance full stack web development projects from requirements to deployment.
- Mentored 20+ students in Mathematics and Science as a freelance tutor, strengthening academic performance up to Class 10.